Hassan Imani

hasanimani.pa@gmail.com | LinkedIn: hassan-imani-parashkooh | +1 (778) 513-1039

PROFESSIONAL EXPERIENCE

Machine Learning Engineer | AML AI Research & Special Projects — Jul 2025 – Present
Scotiabank — *Toronto, ON*

- **Foundational Representation Learning:** Contributed to the self-supervised sequence learning for dense client embeddings; led downstream evaluation and analysis, validating semantic alignment across demographic and behavioral cohorts.
- **Generative Modeling & Anomaly Isolation:** Refactored, stabilized, and optimized the team's anomaly detection frameworks, conditioned on KYC profiles, isolating peer-group anomalies to suppress false positives.
- **ML Platform & Automated HPO:** Developed an Hyperparameter Optimization (HPO) tool to traverse parameter spaces; designed synthetic transactional benchmarks to unblock delayed-feedback validation loops.
- **Distributed Feature & Inference Pipelines:** Implemented end-to-end detection pipelines from distributed SQL/feature extraction through model inference to LLM-enabled alert narratives.

Machine Learning Engineer — Feb 2024 – Jul 2025
Zown — *Toronto, ON*

- **Listing Recommendation Engine:** Architected and deployed an end-to-end matching engine pairing hard-filtering candidate retrieval with a dual-encoder Two-Tower neural network.
- **Multi-Task Engagement Objectives:** Integrated auxiliary multi-task MLP heads predicting dwell time and interaction click-throughs, enriching user/item representations and ranking relevance.
- **Multimodal Content Understanding:** In-housed listing photo understanding by fine-tuning a Vision Transformer (Google ViT) with multi-task classification and regression heads, feeding downstream ranking features.
- **Continuous Training & Valuation Systems:** Deployed deep tabular property valuation models backed by an automated Continuous Training (CT) pipeline and holdout validation gating to guard against production metric degradation

EDUCATION

York University — 2021 – 2025
PhD in Mechanical Engineering — *Toronto, ON*

- Equivariant Graph Neural Networks (eGNNs), domain adaptation via topological priors, spatio-temporal trajectory tracking, and statistical regression modeling.

Tehran Polytechnic — 2018 - 2021
MSc in Mechanical Engineering — *Tehran, IR*

Sharif University of Technology — 2013 - 2018
BSc in Mechanical Engineering — *Tehran, IR*

TECHNICAL SKILLS

Languages & Distributed: Python, SQL, Bash, PySpark, Spark

ML & Architectures: Two-Tower Recommenders, Representation Learning, GNNs, Vision Transformers, Anomaly Detection

Tools & Platforms: PyTorch, PyTorch Geometric, Git, Docker, GCP, MongoDB, Copilot

PUBLICATIONS

Graph Neural Networks for Sustainable Energy (HI Parashkooh, C Jian) — 2025
Predicting Adsorption in Aromatic Molecules — *ChemE 9 (4), 85*

An Innovative Method for Calculating PMV (HI Parashkooh, C Jian) — 2024
from a standard molecular trajectory — *J. Mol. Liq. 395, 123879*

Data Mining Guided Molecular Investigations (HI Parashkooh, C Jian) — 2022
on the Coalescence of Water-in-Oil Droplets — *Energy Fuels 36 (4), 1811*